Datto SIRIS registration workflow

Observed on October 10, 2026

Overview and scope

This document maps the locally observable registration workflow in Datto SIRIS OS 2 5.6.3: page load, the stock network/update behavior, the optional storage pane, the required name/admin/configure/review panes, and final submission. It also records the offline-replica overrides that remove the network/update gate and complete native local registration without a Datto Cloud connection.

The browser calls JSON-RPC methods on the appliance itself. The appliance's PHP services perform the server-side work. Browser behavior is described directly from readable JavaScript and Twig templates. Server behavior is marked as inferred where the implementation is encoded.

What was examined

- **spf-datto-pkgs (10.0.3.200):** the unregistered replica. Most code reading was done there to keep load off the production appliance.
- **spf-datto (10.0.3.147):** the registered appliance. It was used only for comparisons such as file names, feature-flag state, and `upgradectl` behavior.

The two systems had the same `datto-siris-os-2` package version and the same relevant web application layout. That establishes a useful comparison point; it does not prove that every unpackaged file is byte-identical.

Workflow at a glance

Order	Pane or action	Condition	Browser-visible behavior
1	Page load	Always	Server chooses the panes and embeds session/CSRF data.
2	Network check	Stock workflow only	Retries every five seconds until connectivity succeeds.
3	Update	Stock server-selected	Starts an OS upgrade and polls progress once per second.
4	Storage	Server-selected	Inspects drives and creates the selected storage pool.
5	Name	Always	Collects the device name.
6	Administrator	Always	Collects admin credentials and rates password strength.
7	Configure	Always	Loads the default offsite location and collects configuration.
8	Review	Always	Presents the values before submission.
9	Submit	Stock	Calls `send`, whose registration service calls Datto Cloud.

Order	Pane or action	Condition	Browser-visible behavior
10	Offline submit	Air-gapped replica	Calls a same-origin local finalizer, then logs in locally.

```
page load
   |
   v
session + CSRF context
   |
   v
network check ---- retry every 5 s until successful
   |
   +---- optional update ---- attemptUpgrade -> getProgress loop
   |
   +---- optional storage --- checkStorage -> createStorage
   |
   v
name -> administrator -> configure -> review -> send
```

The server decides whether the update and storage panes exist. The browser builds only the panes present in the rendered page. On the air-gapped replica, setup forces the update-pane switch off and suppresses the network poll when no online-only pane remains:

```
offline page load -> name -> administrator -> configure -> review
                  -> local finalizer -> local login
```

Where the code lives

Registration is implemented in `/usr/lib/datto/device` (part of the `datto-siris-os-2` package) and its web root, `/datto/web`. Paths below are relative to `/usr/lib/datto/device` unless they begin with `/`.

Role	File	Readable
Route `/register`	`config/routes.yaml` (`registration`)	Yes
Route `/api`	`config/routes.yaml` (`api`)	Yes
Page controller	`src/App/Controller/Web/Registration/RegisterController.php`	Encoded
Wizard API endpoint	`src/App/Controller/Api/V1/Device/Register.php`	Encoded
Registration services	`src/Service/Registration/RegistrationService.php`, `Registrant.php`	Encoded
Feature framework	`src/Feature/FeatureService.php`, `Context.php`, `DeviceRole.php`, `Features/`	Encoded
Wizard page	`templates/commonlookandfeel/Registration/Register/index.html.twig`	Yes

Role	File	Readable
Wizard panes	`update`, `storage`, `name`, `admin`, `configure`, `review` pane templates	Yes
Wizard logic	`/datto/web/commonlookandfeel/js/App/Registration/RegisterWizard.js`	Yes
Offline final-submit override	`/datto/web/commonlookandfeel/js/App/Registration/OfflineRegister.js`	Yes
Offline local finalizer	`/datto/web/offline-register.php`	Yes
Wizard framework	`/datto/web/commonlookandfeel/js/Widget/Wizard3.js`	Yes
API client	`/datto/web/commonlookandfeel/UI/scripts/classes/Datto/API/Client.js`	Yes
Shell feature library	`/datto/scripts/feature_utils`	Yes
Flag recalculation	`datto-feature-flag-update.service` and `.timer`	Yes
Cloud flag refresh	`datto-cloud-feature-refresh.service` and `.timer`	Yes
OS upgrade tool	`/usr/local/bin/upgradectl`	Root-only binary

Browser-to-device API

Stock wizard actions are JSON-RPC 2.0 POST requests to `/api?<method>` on the appliance. The method is also present in the JSON request body. `RegisterWizard.js` creates the API client, and the client sends `{jsonrpc: "2.0", method, params, ...}` with jQuery.

The page embeds a CSRF token and token identifier. Requests without a valid session and token are rejected with HTTP 403. The update pane uses the loss of the old session during a reboot as one signal while it waits for the appliance to return.

Method	Called when	Purpose
`testNetworkConnection`	Page load, then every five seconds	Gate network-dependent panes.
`attemptUpgrade`	Opening the update pane	Start the OS upgrade.
`getProgress`	Every second during upgrade	Read running and success state.
`checkStorage`	Storage pane	Inspect available drives.
`createStorage`	Storage pane	Create the pool on the selected drive.
`getDefaultLocation`	Configure pane	Load the default offsite datacenter.
`v1/device/security/getPasswordStrength`	Administrator pane	Rate the proposed password.
`send`	Finish	Submit the completed workflow.

On the air-gapped replica, only the finish action differs: `OfflineRegister.js` sends the same collected values as JSON to `/offline-register.php`. That endpoint is local to the appliance, requires a same-origin AJAX POST, and does not expose a general JSON-RPC client.

The wizard also loads `/register?devicetable` with a plain AJAX GET to render the storage pane's drive table.

What happens when the page loads

1. **The server renders the page.** The encoded controller supplies switches such as `updatePaneEnabled` and `storagePaneEnabled`, plus `showDatacenters`, `storageCapacity`, session/CSRF values, and navigation URLs.
2. **The template chooses the panes.** Update and storage are added only when their server-provided switches are true. Name, administrator, configure, and review are always added.
3. **The browser builds the wizard.** `RegisterWizard` inherits from `Wizard3` and attaches validation and load/unload hooks to each pane.
4. **The stock network check starts immediately.** A successful response hides the warning and sets `networkConnected = true`. Other results show the warning and schedule a retry five seconds later. The stock update pane cannot advance while the flag is false.

The rules that select optional panes are server-side and encoded. Their exact conditions should not be inferred solely from variable names.

Update behavior

Opening the update pane calls `attemptUpgrade`. While an upgrade is active, the browser calls `getProgress` once per second and reads `isUpgradeRunning` and `wasUpgradeSuccessful`. A reboot invalidates the original web session, so transient HTTP 403 responses are expected while the appliance restarts.

The browser-side polling behavior is confirmed. Package selection, rollback behavior, and the internal actions performed by `/usr/local/bin/upgradectl` remain server-side or binary implementation details and require separate observation.

Offline replica override

The air-gapped setup modifies both registration templates so `updatePaneEnabled = false` before the pane list and browser context are built. It also changes `RegisterWizard` to call `checkNetworkConnectivity()` only when an authorization or update pane is present. With both online-only panes absent, the wizard sets `networkConnected = true` locally and begins at **Name the Device**.

This override fixes an inconsistency in the stock failure path: `setUpgradeError()` visually enables **Next**, but the update pane's validator still rejects the click while `networkConnected === false`. Production Twig caches are invalidated during setup so the offline template takes effect immediately.

The stock final `send` action still fails offline because the encoded `RegistrationService` first calls Datto Cloud method `v1/device/registration/registerNow`. The air-gapped setup therefore installs a narrow, one-time

local finalizer and a browser override for the finish action:

1. The endpoint accepts only POST, verifies the AJAX header and same-host referrer, validates the submitted values, and takes a nonblocking registration lock.
2. It boots the installed Datto application and obtains the native `RegistrationService` and `Registrant` classes.
3. For this call only, it replaces every cloud client reachable from the registration service with one in-process client. The four observed methods are explicitly allowed: `v1/device/registration/registerNow`, `v1/device/contactInformation/setDeviceContactInformation`, `v1/device/billing/getServiceInfo`, and `v1/device/ddns/update`. The billing method receives an empty local result; the other three receive local success. Any different method is rejected, and the substitute opens no network connection.
4. It calls the native `register` routine, preserving the local hostname, administrator, timezone, and registration-marker behavior. It then explicitly enables `LocalLoginService` so the appliance accepts the new local administrator instead of requiring Partner Portal authentication, and verifies the native `isRegistered()` result.
5. A pending marker lets a retry remove only an administrator created by the same interrupted registration attempt. Successful registration removes that marker and any stale `reg.error` file.
6. The browser performs the local login and redirect after the endpoint succeeds.

This is local appliance onboarding, not Datto Cloud account enrollment. Licensing, updates, remote management, and other cloud services are not emulated and remain unavailable without connectivity.

Storage and final configuration

The storage pane calls `checkStorage` to populate the device table and `createStorage` for the selected drive. The later panes collect the device name, administrator settings, offsite location, and remaining configuration. The review pane presents the completed values before the stock `send` action or the offline local finalizer submits them.

The encoded native service was observed to begin its local work only after the cloud `registerNow` result. It writes the registration marker before requesting billing data and later updates DDNS. The retry marker protects against failures between user creation and final completion. Other persistence details remain encoded and should be evaluated through logs and before/after state, not inferred from the browser sequence alone.

Feature flags

The installation contains roughly 154 feature classes, plus local recalculation and cloud-refresh timers. The readable shell helper exposes flag values to scripts, while the PHP service determines flags for the web application.

This document records only directly observed flag state. A useful follow-up is a registered-versus-unregistered comparison containing the flag name, both values, source, refresh mechanism, and observed effect.

Evidence boundaries and open questions

- Browser request order, pane construction, polling intervals, and client-side gates are directly observable.
- PHP controller and service internals are encoded; all four registration-time cloud methods and subsequent native local actions were confirmed through a snapshot-protected runtime trace, while exact persistence details still require before/after observation.
- The exact destination tested by `testNetworkConnection` was not established from the browser code alone.
- The conditions controlling the update and storage panes need controlled-state testing.
- Upgrade rollback and interrupted-storage behavior still need explicit failure tests.